

“Yeah, Right...”

Sarcasm Classification on Social Media

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Introduction

Social behavior in humans are a complex subject that can seem almost impossible to fully understand. A good example of this are usage of sarcasm in interactions, which is famously easily to create misunderstandings, especially when written, because of intonation and voice features are often used as markers for a sarcastic comments. Combined with a internet where billions of conversations take place everyday about the most obscure topics makes this a complicated and nuanced classification problem.

This creates the interesting question: How well can the machines, which we run these websites on, understand what is really being said? Therefore this project aims to explore how well different machine learning models can classify sarcasm from sentences made on social media.

Methodology

Models:

- **Logistic Regression** – To create a good benchmark for the project
 - Size: 304 parameters
- **BiLSTM with multi-head attention** – To explore the connection of different attention and sarcasm
 - Size: 474 629 parameters
- **BERTweet**– Showcase the power of fine-tuning pre-trained networks
 - Size: 134 900 737 parameters

Data:

- Dataset containing 1 000 000 sentences from social media platform “Reddit.com” with associated label 0 for normal comment and label 1 for sarcastic.
- Created by scraping the website, where comments with the tag \s was labeled sarcastic, and otherwise normal. Could lead to noise.

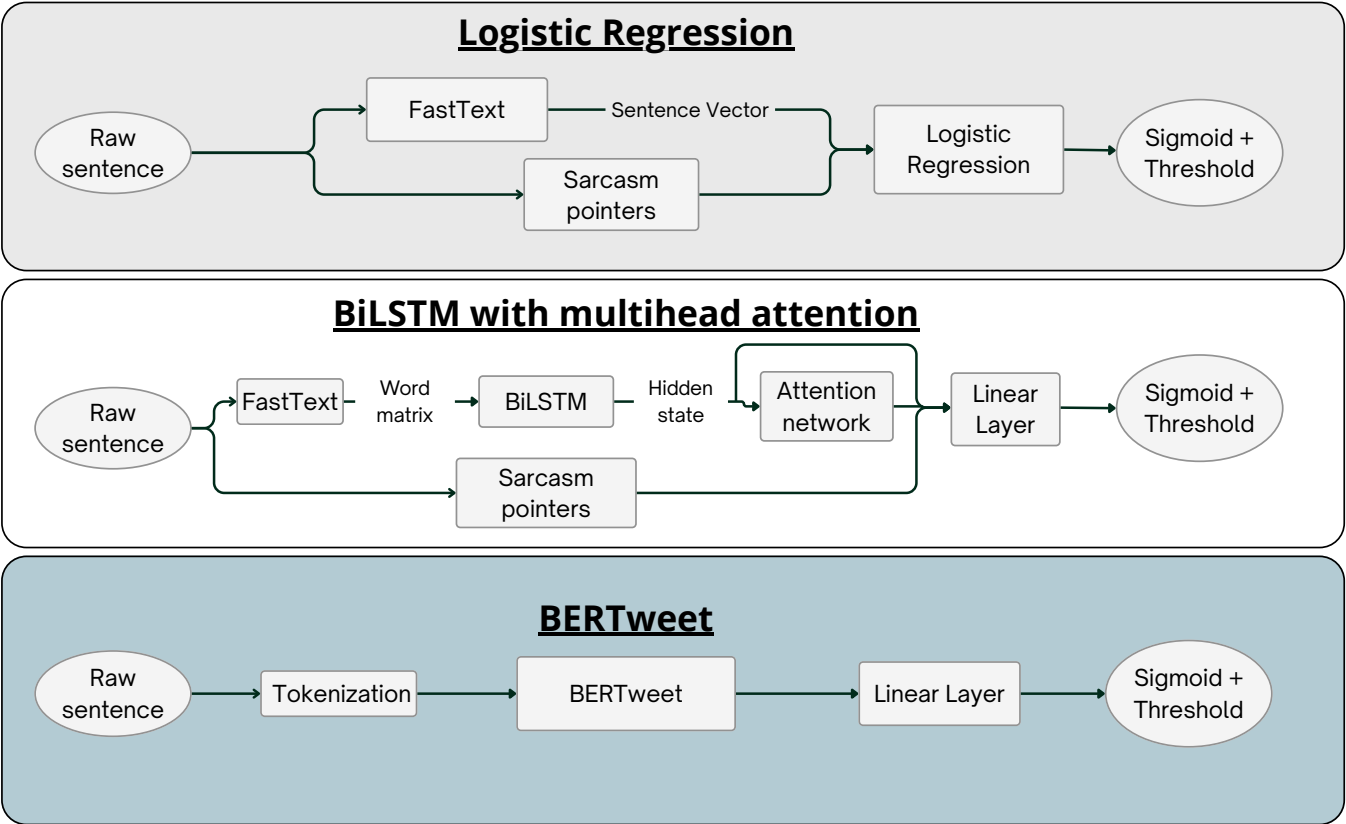


Figure 1: Flow chart over the different models used in the project

Results

- Logistic regression almost always guesses sarcastic, but better accuracy than picking randomly.
- BiLSTM performed better with a more even distribution between the the guessing, but still favored classifying sarcastic.
- BERTwitter preformed the best, and had the most evenly distributed guesses.

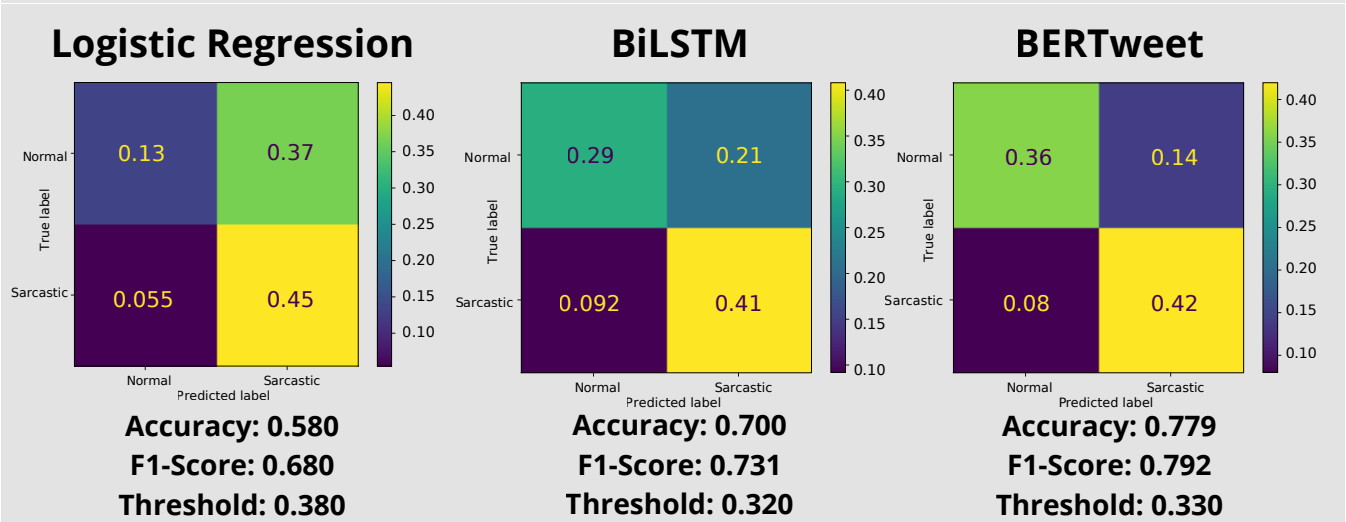


Figure 2: Confusion matrix and some performance metrics over the models

Discussion & Conclusion

- It's clear that it's possible for these models to classify sarcasm. Because of the missing context of the sentences they all in some way succeed.
- To continue this work, an improvement would be to some way input the comment that the sentence answered to, for the model to see the full picture.
- The threshold is quite low for all of the models. Is this because of the sentences being easily interpreted as sarcastic without context, or something else?

